

Felix Omondi & Ezekiel Ekanem

CMPU-395-51: Machine Learning (Spring 2026)

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Project Progress Report

GitHub Link: <https://github.com/Felomondi/AI-and-future-of-work/tree/main>

AI, Gender, and the Future of Work: Predicting Occupational Exposure to Artificial Intelligence

1. Further Data Preparation Since Proposal

Since submitting our initial project proposal, we have finalized our data wrangling and cleaning pipeline, which is fully documented in our *data_cleaning.ipynb* notebook. The primary challenge was integrating the disparate datasets: the O*NET task/skill data, the AI Occupational Exposure (AIOE) scores, and the BLS Current Population Survey (CPS) demographic data.

To prepare the dataset for predictive modeling, we pivoted the ONET files from a long to a wide format, standardizing all 8-digit SOC codes to a 6-digit format to successfully merge them with the BLS OEWS and CPS files. During this process, we instituted a strict threshold, dropping any columns with more than 20% missing values to preserve data integrity, while explicitly protecting our target variables (AIOE, LMOE, IGOE) and critical demographic predictors (like the percentage of women and Black workers). We imputed the remaining missing values using column medians. Finally, we applied a standard scaler to the ONET predictor variables to ensure our regularized models process them correctly. Our final, clean dataset contains 631 occupations and 650 variables with zero missing values.

2. Documentation of Analyses and Preliminary Results

With the data cleaned, we transitioned into coding the modeling framework for Research Question 1 (RQ1), which focuses on predicting AI exposure based on O*NET characteristics. We set up a Google Colab notebook to implement a 5-fold cross-validation strategy paired with randomized search for hyperparameter tuning.

Failed/Unexpected Analyses: Our initial unregularized OLS baseline unexpectedly yielded a near-perfect R^2 (0.98), which we quickly diagnosed as data leakage caused by leaving alternative AI exposure indices (LMOE and IGOE) in the feature set. After strictly isolating our predictors and removing LMOE and IGOE from the feature set, the corrected OLS baseline achieved a more realistic test R^2 of 0.38. This means that our O*NET task and ability features can explain roughly 38% of the variance in a job's AI exposure. This low number is as a result of the high dimensional feature space (approx. 640 variables), which exceeds our training sample sizes as well as multicollinearity between the variables. This outcome definitively validates our

methodological need for automated feature selection and regularization via Lasso and Ridge regression to handle the inherent multicollinearity and high dimensionality.

Completed Analyses and Preliminary Results: To correct the multicollinearity and dimensionality issues, we successfully implemented **Lasso (L1)** and **Ridge (L2)** regressions.

- **Lasso Regression (L1):** Our preliminary runs confirm that Lasso successfully resolved our extreme multicollinearity problem by acting as a strict automated feature selector. Out of approximately 640 variables, the model shrank 567 redundant O*NET features to exactly zero. This left a highly stable, interpretable model utilizing only 71 core predictors, achieving a robust test R^2 of 0.975. Analyzing these remaining non-zero coefficients provided a clear narrative of what drives AI exposure: cognitive skills (e.g., written comprehension and memorization) emerged as the primary risk factors, while physical abilities (e.g., gross body coordination and stamina) acted as strong protective factors against automation.
- **Non-Linear Ensembles:** To capture complex feature interactions that linear models might overlook, we implemented and tuned Random Forest and XGBoost regressors. Both models demonstrated exceptional predictive power, with Random Forest achieving a cross-validated R^2 of 0.961 by utilizing deep decision trees. However, XGBoost emerged as our highest-performing model overall, achieving an R^2 of 0.970 and a remarkably low Mean Absolute Error (MAE) of 0.121. XGBoost achieved this superior accuracy by taking the opposite structural approach: methodically building shallow, "weak" trees (max depth of 3) and utilizing a slow learning rate to sequentially correct past errors. We thereafter used this optimized XGBoost model as our primary engine for extracting SHAP feature interpretations.
- **Interpretability (SHAP):** We applied SHAP (Shapley Additive Explanations) to unpack our optimized XGBoost model, and the resulting visualizations definitively confirmed our hypothesis. The SHAP summary plots reveal that cognitive traits related to text processing (e.g., Written Comprehension and Expression) act as primary risk factors, driving AI exposure scores upward. Conversely, physical attributes (most notably Extent Flexibility, Multilimb Coordination, and Strength) act as powerful protective shields against automation. Furthermore, SHAP dependence plots uncovered a distinct, non-linear relationship: an occupation's AI exposure drops dramatically the moment it requires moderate-to-high physical flexibility, perfectly illustrating the boundaries of current AI capabilities in the physical world.
- **Manifold Learning (UMAP):** We executed a UMAP projection to compress the 640-dimensional O*NET task space into two dimensions, revealing natural groupings of occupations based solely on their task similarities. When overlaying the AIOE scores as a color map, a striking gradient emerged: high-exposure occupations cleanly separated from low-exposure ones. Because the UMAP algorithm was entirely blind to the target variable during projection, this distinct clustering visually validates our machine learning

results. It proves that an occupation's core structural profile - specifically the divide between cognitive and physical labor - fundamentally dictates its vulnerability to AI automation.

3. Changes in Research Questions, Methods, or Topic

At this stage, our core research questions and topic remain unchanged. However, based on the success of the XGBoost model for RQ1, we are slightly refining our approach for RQ2. Instead of relying solely on linear Double Machine Learning (DML), we are investigating whether we can use non-linear tree-based models as the auxiliary predictors within the DML framework to better isolate the causal impact of demographic composition on AI exposure.

4. Writing Progress

Since the proposal, we have drafted the formal "Data cleaning" sections of our final report, translating the steps logged in our *data_cleaning.ipynb* file into academic prose. We have also written out the "Methods" subsections detailing the mathematics and justification behind our use of Lasso, Ridge, and XGBoost for RQ1.

5. Detailed Plan for Rough Draft (Due Friday, April 24)

To ensure we have a complete rough draft ready for peer review by April 24, we have established the following milestones:

- Expand RQ1 by exploring LMOE and IGOE target variables to determine variation in skills that are more susceptible to be replaced by language models and image generation models
- Explore Research Question 2
- Write the "Results" section of the paper, systematically addressing RQ1 and RQ2. Integrate all tables and figures in the paper.
- Draft the "Discussion" section, linking our findings back to the Frey & Osborne (2017) and ILO (2025) literature. Review the entire draft for flow, clarity, and formatting consistency.
- Final proofread and submission of the rough draft.